Chapter – 9
Environment Management in SAIL

9.1 Introduction:

From times immemorial, steel has played a crucial role in the development of societies as well as nations at large, and continues to be the backbone of the modern industrial society. No material yet invented by man can match the versatility of steel in terms of strength, availability, durability, formability, affordability, recyclability and cost. Steel is at the core of a green economy, in which economic growth and environmental responsibility work hand in hand. As a product, it is the most eco-friendly amongst all the materials. Once steel is produced, it becomes a permanent resource because it is 100% recyclable without loss of quality and has a potentially endless life cycle. Steelmaking processes and related activities consume materials and resources and generate significant amount of emission, effluent and wastes. Air, water and land are affected due to the environmental impacts of the steelmaking operations.

The sustainable consumption of resources by employing efficient processes and activities helps in alleviating the overall environmental impacts of an industry by reducing wastage of resources, thereby mitigating the production of wastes, emission and effluent. Pollution control measures ensure that wastes, emission and effluent are properly treated and disposed in a manner to have none or minimal environmental impacts at the end of the process. Key environmental concerns for the iron and steel industry are the emission of air pollutants, effluent, production of greenhouse gases and the management of solid as well as hazardous wastes.

SAIL lays a strong emphasis on environment impacts and its mitigation, along with production and profitability and considers clean environment practice at the core of its every industrial activity. The **Corporate Environmental Policy** emphasizes conducting our operations in an environmentally responsible manner to comply with applicable regulations and striving to go beyond.

Corporate Sustainability Policy 2026

Steel Authority of India Limited (SAIL) remains committed towards reducing its environmental footprint, enhancing its social impact and securing long-term economic growth, all while fostering a more sustainable future for its stakeholders. The objectives will be achieved through well-defined policies and initiatives which on a broader perspective are drawn from and are in congruence with SAIL's overall Vision and CREDO. SAIL reaffirms its commitment towards a sustainable future through:

- Environmental Stewardship: Commitment to fostering a clean and sustainable environment by continually enhancing environmental performance through initiatives such as resource optimization, emission reduction, waste management programs, biodiversity conservation, energy efficiency projects, digitalization and adoption of best available technologies, with a progressive pathway towards decarbonization and circular economy.

- Social Responsibility: Creating of a positive footprint within the society to make a meaningful difference in the lives of people by continually aligning its initiatives to the cornerstones of social sustainability & health, rights and equal opportunities, community engagement and by adopting strongest safety standards and reducing workplace incidents.
- Ethical Governance: Striving to conduct business with integrity, in an ethical and transparent manner, to demonstrate its commitment towards meeting the interests of SAIL's stakeholders through various Corporate Governance mechanisms.
- Economic Sustainability: Maintaining commitment to long-term business viability by embracing innovation and technology, ensuring quality control, practicing effective risk management, fostering continuous improvement and investing in capacity building and consultative feedback and review.
- Sustainable Sourcing: Ensuring that the procurement of goods and services is responsible, safe and sustainable.

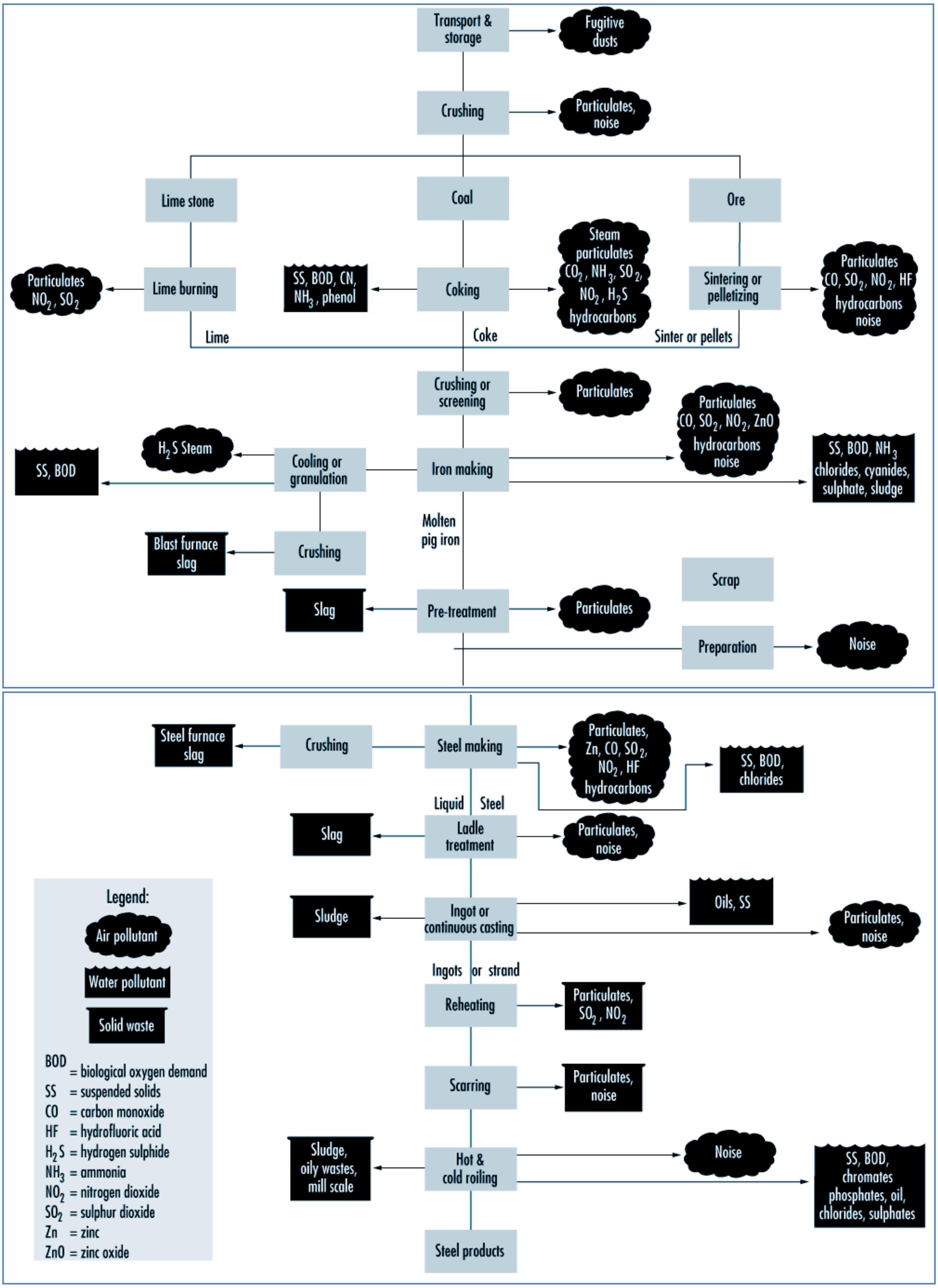
Environmental concern associated with steelmaking

The production of steel in an integrated steel plant involves several operations, starting from use of raw materials like iron ore, coal and flux in production of hot metal and further processing of hot metal into steel and subsequently, rolling of steel into finished products in the rolling mills.

Several environmental pollutants are released in an integrated steel plant during the various processes from raw materials to finished products. Emissions such as carbon dioxide (CO_2), sulphur oxides (SO_x), nitrogen oxides (NO_x) and dust are generated due to use of energy and raw materials in the steelmaking process. Water is used throughout the plant for cooling, heat transfer, descaling, dust scrubbing, quenching and other processes, thus converting to wastewater after use. One more factor which affects the environment is the noise which is produced during the operations of plant machineries. During the course of steelmaking process, many solid wastes are generated which include blast furnace (BF) slag, BF flue dust, BF sludge, BOF slag, BOF sludge, mill scale, waste refractory bricks etc. BF slag and BOF slag contribute the major share of solid wastes generation in the steel industry.

In short, there are four types of pollutants, viz. air emissions (primary and secondary emissions), effluent discharges, generation of noise and solid waste. The generation quantity of various types of pollutant differ from one steel plant to other depending upon the steelmaking processes adopted and pollution control equipment installed.

Different pollutants and sources of pollution in an integrated steel plant are depicted below:


Transport & storage
Fugitive dusts
Crushing
Particulates, noise
Lime stone
Coal
Ore
Particulates NO_2, SO_2
Lime burning
SS, BOD, CN, NH_3, phenol
Coking
Steam particulates CO_2, NH_3, SO_2, NO_2, H_2S hydrocarbons
Sintering or pelletizing
Particulates CO, SO_2, NO_2, HF hydrocarbons noise
Lime
Coke
Sinter or pellets
Crushing or screening
Particulates
Particulates CO, SO_2, NO_2, ZnO hydrocarbons noise
SS, BOD
H_2S Steam
Cooling or granulation
Iron making
SS, BOD, NH_3 chlorides, cyanides, sulphate, sludge
Molten pig iron
Blast furnace slag
Crushing
Scrap
Slag
Pre-treatment
Particulates
Preparation
Noise
Steel furnace slag
Crushing
Steel making
Particulates, Zn, CO, SO_2, NO_2, HF hydrocarbons
SS, BOD, chlorides
Liquid Steel
Slag
Ladle treatment
Particulates, noise
Legend:
Air pollutant
Water pollutant
Solid waste
BOD = biological oxygen demand
SS = suspended solids
CO = carbon monoxide
HF = hydrofluoric acid
H_2S = hydrogen sulphide
NH_3 = ammonia
NO_2 = nitrogen dioxide
SO_2 = sulphur dioxide
Zn = zinc
ZnO = zinc oxide
Sludge
Ingot or continuous casting
Oils, SS
Particulates, noise
Ingots or strand
Reheating
Particulates, SO_2, NO_2
Scarring
Particulates, noise
Sludge, oily wastes, mill scale
Hot & cold roiling
Noise
SS, BOD, chromates phosphates, oil, chlorides, sulphates
Steel products

9.2 Environment Management

As an environmentally conscious and responsible company, SAIL takes its responsibility sincerely towards protection of the environment both at its plants & mines and the community in which it operates, and is dedicated towards the practices of environmental techniques/procedures to achieve sustainable development. Environment management at SAIL is a multi-layered process. All environmental programmes at SAIL plants/units and mines address critical issues such as resource optimisation, environment protection, mitigation of adverse environmental impacts, if any, as well as smooth operational and maintenance programmes for environment and pollution control devices and facilities.

SAIL established the corporate unit "Environment Management Division" (EMD), headquartered at Kolkata, on 23rd June, 1988, for sound management of environment at plants/units and mines of the company as well as promote sustainable environment management practices across SAIL. EMD has its unit office at Delhi. EMD is certified with Quality Management System (QMS) linked with ISO 9001:2015.

The primary role of EMD is to apprise the SAIL management regarding the environment and pollution control activities at plants/units and mines located across India. It also plays a pivotal role in consolidation of the efforts of the plants/units and mines towards environment protection and resource optimization through its multifarious activities involving

- proactive interface between SAIL units and the regulatory agencies,
- monitoring and assessment,
- technology dissemination and
- awareness campaigns etc.

Salient activities of EMD are outlined below:

- Coordination with the plants/units and mines for environmental compliance
- Development of Management Information System (MIS)
- Environmental appraisal of project proposals
- Sustainability Reporting, Managing ESG Risks, Sustainable Development (SD) Projects, Development of Corporate Environmental Vision/Policy and Action Plans
- Reporting of deviations w.r.t. Environmental Rules & Regulations, Directions issued by different regulatory authorities to SAIL Board through Board Sub-committee on Health, Safety & Environment
- Facilitating both SAIL and individual units for participation in various environment related awards
- Review of environmental performance of SAIL plants/units and mines
- Facilitating SAIL plants/units and mines for the following activities:
  - Obtaining statutory clearances like Environmental Clearance (EC), Consent to Operate (CtO), Authorisation etc.
  - Solid Waste Management, Water Conservation, Plantation
  - Addressing Climate Change Issues, Life Cycle Inventory & Life Cycle Assessment (LCA)
  - Liaisoning with State/Central statutory authorities/ministries
  - Implementation of Environment Management Systems: EMS (ISO 14001)

- Organising workshops/ training programmes
- Coordination with World Steel Association (WSA), Indian Steel Association (ISA), & other Indian Industrial bodies/associations.
- Compiling environmental data/information for replies to the queries raised in the Parliament

Along with corporate EMD, Environment Control Department (ECD) was also set up at individual plant to closely monitor the environmental performance of the plants and take care of the statutory environmental requirements at plants. Similarly, Environment Cell had also been established at mine level to look after the environmental activities at the mines. Major functions of these departments are:

- Monitor and report the quality of emission from the stacks, effluents and ambient air in and around plants as per the guidelines issued by statutory agencies, viz. Ministry of Environment Forests & Climate Change (MoEFCC), Central Pollution Control Board (CPCB), State Pollution Control Board (SPCB), etc.
- Obtain clearances/consents/authorisation as per regulatory requirements.
- Submit compliance reports to statutory bodies as per requirement of various rules/regulations.
- Coordinate with statutory authorities of state on environmental issues for smooth functioning of plants.
- Prepare action plans for improving the environmental performance of the plant and implement the same.

Compliance of Statutory Requirements:

The Iron and Steel industry in India is guided by the following principal environmental acts and rules:

- **The Water (Prevention and Control of Pollution) Act, 1974**
  - The Water (Prevention and Control of Pollution) Rules, 1975
- **The Air (Prevention and Control of Pollution) Act, 1981**
  - The Air (Prevention and Control of Pollution) Rules, 1982
- **The Environment (Protection) Act, 1986**
  - Environmental (Protection) Rules, 1986
  - Noise Pollution (Regulation & Control) Rules, 2000
  - Environmental Impact Assessment (EIA) Notification, 2006
  - Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016
  - Bio-medical Waste Management Rules, 2016
  - Solid Waste Management Rules, 2016
  - Plastic Waste Management Rules, 2016
  - Construction and Demolition Waste Management Rules, 2016
  - E-Waste Management Rules, 2022
  - Battery Waste Management Rules, 2022

SAIL, as a responsible Corporate Organisation, takes care of environment right from the very first day of construction of any new project/activity and during the course of its operation as well.

SAIL plants/units and mines obtain prior Environmental Clearance (EC) in case of any new project and expansion/modernization of the existing project in line with the EIA Notification, dated 14th September, 2006.

Similarly, as per the provisions of the Water (Prevention and Control of Pollution) Act, 1974 and the Air (Prevention and Control of Pollution) Act, 1981, Consent to Establish (CtE) before establishment of a new project and Consent to Operate (CtO) are regularly obtained from the concerned state pollution control board prior to start of its operation.

Authorisations for handling and management of hazardous waste and bio-medical waste are also obtained as per the requirement of the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 and the Bio-Medical Waste Management Rules, 2016 respectively.

9.3 Pollution in Steel Plants

Control of Air Pollution:

Starting from handling of raw materials to production of finished steel, air pollution control equipment/facilities like Bag Filters, Electro Static Precipitators, Multi Cyclones, Venturi Scrubbers, De-dusting Systems, De-fuming Systems, and Covered conveying Systems etc. are installed in all the pollution-prone areas to control Particulate Matter (PM) emission.

Emission of SO_2 is controlled through use of coal with low sulphur content and desulphurized coke oven gas as fuel. For abatement of NO_x emission, specially designed (multi-slit) ignition burners are installed along with optimised process parameters.

During the modernisation-cum-expansion programme (MEP), various state-of-the-art energy-efficient technologies/facilities such as Coke Dry Quenching (CDQ) with tall Coke Oven Batteries, Coal Dust Injection (CDI), Top gas Pressure Recovery Turbine (TRT) & Waste Heat Recovery (WHR) from hot stoves of Blast Furnaces, Waste Heat Recovery (WHR) from sinter coolers of Sinter Plants, Continuous Casting, Walking Beam Reheating Furnace equipped with Rolling Mills, Torpedo Ladle for hot metal handling are installed in all the plants. Apart from these, by-product gases are utilised for captive generation of power.

Control of Water Pollution:

In order to preserve water quality of natural water bodies, Effluent Treatment Plants (ETPs) are provided with each individual shop of all the plants to treat and recirculate the industrial waste water for further use. Moreover, Sewage Treatment Plants (STPs) have also been installed in the steel townships and plants to treat the sewage for achieving the stipulated standards.

Over and above, comprehensive water conservation schemes for treating and recycling waste water are being implemented to achieve long term goal of Zero Liquid Discharge (ZLD).

Besides, revamping/upgrading of localized recirculation systems and water auditing by third party, installation of rainwater harvesting schemes is one such environment-friendly initiative towards water conservation. SAIL has been implementing Rain Water Harvesting (RWH) systems and artificial recharge to ground water since long and adopted it as an integral part of water conservation measures. Considering the factors like geographic conditions of the site, regulatory guidelines and availability of infrastructure facilities, rain water harvesting systems are being implemented with utmost care. Further, in consonance with 'Catch the Rain' campaign of Ministry of Jal Shakti, rain water harvesting facility is envisaged during conceptualization of any upcoming project and is incorporated in the technical specifications of the proposal.

Control of Noise Pollution:

Various noise controlling measures like acoustic enclosures, hoods, acoustic lagging are adopted to reduce noise level at sources like high speed machineries viz., compressors, fans and blowers. Induced Draught (ID) and Forced Draught (FD) fans are properly equipped with silencers and insulated casing. Noise-proof and air-conditioned control rooms are provided for the operators wherever required.

Management of Wastes:

SAIL is committed in its Corporate Environmental Policy to reduce solid waste generation and maximise its utilisation to achieve 100% and follows the principle of 4R's (Reduce, Reuse, Recycle and Recover) in the area of solid waste management to be sustainable in the steel sector.

The molten BF slag is granulated through Cast House Slag Granulation Plants (CHSGPs) installed at BFs and is sold to cement industries for its consumption as an input material. BOF slag containing iron bearing particles is processed and the metallic part from the slag lumps is separated out before its recycling back to the process. BOF slag is used either in blast furnace as a replacement of limestone, or in sinter making through base-mix. The other wastes such as BF flue dust, mill scales, lime/dolo fines, refractory waste etc. either reused fully in the process or sold to the external agencies for its further use.

With an aim to achieve the maximum benefit from the concept of "Waste to Wealth", various R&D based studies have been taken up either through in-house research wing or in association with other eminent research centres or academies of national repute to explore potential avenues for enhancing BOF slag utilization.

SAIL as an industry partner has participated in the Ministry of Steel's sponsored R&D project "Development of steel slag based cost effective eco-friendly fertilizers for sustainable agriculture and inclusive growth" through ICAR-IARI.

SAIL plants have been extensively using steel slag for making internal roads since long. SAIL has also taken steps for utilisation of BOF slag in rural road construction under Pradhan Mantri Gramin Sadak Yojna (PMGSY).

Apart from the above mentioned solid wastes, some of the wastes like used/spent oil, benzol acid tar sludge, decanter sludge/tarry waste, ETP sludge etc. generated during iron and steel making process are hazardous in nature. Utmost care and effective steps for safe handling, transportation and disposal of these wastes are taken at SAIL. These wastes are safely disposed either in the captive Secured Landfill Facility (SLF) or through the authorized handling agency. Some hazardous wastes are reused/co-processed.

Promotion of Renewable Energy Sources:

SAIL is taking initiatives to accelerate its transition toward renewable energy, reinforcing its commitment to sustainable operations across its Plants, Units and Mines.

As a part of this initiative, SAIL has successfully installed 27.8 MW of solar energy capacity, contributing to cleaner and more efficient power generation. Expanding its green energy footprint, solar water heating and solar lighting systems have been integrated into most guest houses and hospitals, further promoting energy conservation in key facilities.

Diversifying Renewable Energy Sources: To further strengthen its sustainable energy mix, SAIL is procuring approximately 7 MW of bagasse based co-generated power at Salem Steel Plant during the sugarcane crushing season, leveraging biomass as a sustainable alternative.

In addition to this, SAIL is increasing its reliance on green power procurement from utilities, reinforcing its commitment to eco-friendly energy solutions covered in detail in the Power section. In FY 25, 44.1 MW green power (other than captive RE production) was sourced at ISP, DSP & RSP. In a significant move toward large scale green power adoption, DSP & ISP have secured arrangements for sourcing 50% of their total grid power requirements from DVC’s Green Power supply. This strategic shift besides reducing dependence on conventional energy, also strengthens the Company’s contribution to India’s clean energy goals.

Looking ahead, SAIL has outlined ambitious plans to scale up its renewable energy capacity to 384 MW by the year 2028- 29, thereby driving long-term environmental sustainability and energy efficiency across its operations. Through these proactive measures, SAIL remains steadfast in its mission to integrate innovative renewable energy solutions, ensuring a greener and more sustainable future for the steel industry.

Switching over to LED Illuminating Systems:

SAIL plants/units are gradually shifting to more energy-efficient and durable LED lighting system, in consonance with the Government of India’s initiative “Unnat Jyoti by Affordable LEDs for All (UJALA) Scheme”.

More than 1.2 million conventional lights have now been replaced with LED lighting systems, including 1.72 lakh new installations in FY 2024–25, underlining our commitment to energy conservation. SAIL has an ambitious target for complete replacement of the conventional lights.

Developing Greenery:

Structured plantation programmes are carried out every year in all the SAIL plants/units and mines depending on availability and prevalence of local species, local soil characteristics and prevailing meteorological conditions. SAIL planted 3 lakh saplings during the year 2024-25, taking our cumulative tally to an impressive 22.4 million saplings.

Biodiversity forms the basis of human survival on earth. Living resources (plants animals and microbes) and their habitats form an integral component of the biodiversity. Zoological and botanical parks are being maintained in townships of SAIL plants for preservation of several species of flora and fauna.

VASUNDHARA - a step towards Biodiversity Conservation and Ecosystem Management:
Vasundhara Biodiversity Park in an area covering 409 acres of land has been developed near the township at Durgapur Steel Plant with water body and development of flora and fauna of local species to address the concerns for ecology, biodiversity and environment management. The local people living in peripheral villages, including residents of the Steel Township, are the beneficiaries in respect of the visual aesthetics, the greenery with a clean & calm environment away from the harsh noises of daily city life.

Eco-restoration of mined out areas:
Restoration and rehabilitation of degraded ecosystem is essential for maintaining and enhancing biodiversity as well as replenishing the ecosystem services.

- Kiriburu-Iron Ore Mines, Jharkhand: During FY 2024–25, an area of 1.5 ha was restored at Kiriburu Iron Ore Mines. Cumulatively, 8.0 ha of mined-out land has been reclaimed and rehabilitated. In partnership with IFP Ranchi, 2,562 saplings were planted under gap-filling and replacement activities during the year.
- Meghahatuburu Iron Ore Mines, Jharkhand: In FY 2024–25, plantation was undertaken over 5.0 ha of backfilled area, bringing the total rehabilitated area to 15.2 ha. Key initiatives include: • Installation of a 20 kW on-grid rooftop solar power plant at Guest House-II, the first of its kind in JGOM, BSL. • Development of a fruit orchard over 0.5 ha near Meena Bazar. • Plantation on World Environment Day 2024 featured fruit-bearing species (Amla, Jackfruit, Jamun) and ornamental plants (Bottle Palm, Areca Palm). • Construction of three catchment area treatment (CAT)/soil erosion control structures near the loading section to mitigate runoff-related erosion.
- Bolani Ore Mines, Odisha: Approximately 4.0 ha of area was restored or rehabilitated in FY 2024–25, contributing to a cumulative total of 219.0 ha
- Barsua Taldih Kalta Amalgamated Lease, Odisha: In FY 2024–25, 5,100 saplings were planted under gap-filling within the mining lease. Backfilling activities have cumulatively covered 7.8 ha to date.
- Purnapani Limestone Mines, Odisha: Purnapani Limestone & Dolomite Quarry (ML-153), spread across 230.525 ha, was granted to Rourkela Steel Plant (RSP) of SAIL in 1960. Mining was suspended on 01.03.2004 due to unsuitability of limestone for steelmaking. A 200-acre degraded site was

successfully restored into a native tropical forest ecosystem, featuring diverse plant communities including grasslands, bamboo thickets, and broad-leaved forests.

- Pandridalli and Rajhara Hills Lease, Chhattisgarh: 24.48 ha area has been rehabilitated through dense plantation.
- Rajhara Hills Lease, Chhattisgarh: 9 ha area has been restored through stabilization of Dump through Geo-Textile Coir matting in part of inactive waste dump of Dalli Mechanised mines.
- Dalli Forest Range, Chhattisgarh: 8.7 ha area has been rehabilitated through dense plantation.
- Mahamaya Dulki Lease, Chhattisgarh: 2 ha area has been rehabilitated through dense plantation.

Environment-friendly disposal of Polychlorinated Biphenyls:
This project being the first of its kind in the country and in compliance with the "Stockholm Convention" on Persistent Organic Pollutants (POPs) was taken up by SAIL at Bhilai Steel Plant in partnership with MoEFCC and UNIDO.

The Stockholm Convention is a global treaty to protect human health and the environment from POPs. Use of Poly-Chlorinated Biphenyls (PCB), a POP in any form shall be completely prohibited by 31^{st} December, 2025. PCBs are used in transformers as dielectric fluid and also as cleaning solvent.

The "National PCBs Disposal Implementation Plan" being administrated through the MoEFCC, Govt. of India, focuses on the reduction and elimination of PCBs in the industrial sector of India.

Environment Management System (EMS):
Environmental Management System (EMS) linked with ISO: 14001 is a voluntary approach to manage the immediate and long-term environmental impacts of an organisation's products, services and processes. SAIL has been a torch-bearer in the establishment of the EMS in the steel industry in our country. In mid-90's, SAIL started implementation of EMS ISO: 14001 in its Salem Steel Plant. Presently, all the integrated steel plants, major units and warehouses of SAIL are compliant with EMS ISO: 14001 Standard. Implementation of EMS has helped SAIL plants and units to ensure that their performance is always well within the applicable regulatory requirements.

9.4 ESG Concerns & Green Steel

Environmental Social & Governance (ESG) initiatives have become a strategic imperative for nearly all organizations over the past year. Increased focus and pressure from investors, regulators, employees and other stakeholders make ESG a topic that is not only critical at the board level, but also essential to cascade throughout organizations operationally.

Adopting ESG principles means that corporate strategy focuses on the three pillars of the environment, social, and governance. This means taking measures to lower pollution, $CO2$ output, and reduce waste. It also means having a diverse and inclusive workforce, at the entry-level and all the way up to the board of directors. ESG may be costly and time-consuming to undertake, but can also be rewarding into the future for those that carry it through.

The Emergence of ESG Clearly, one of the major risks organizations face today is related to the environment. In 2019, the World Economic Forum's Risk Report listed weather events, climate change and natural disasters as the top risks by likelihood and impact. That has not changed. However, several other risks have surfaced based on the extraordinary events of these last two years.

Social factors have become front and center in many discussions. Economic disparities, the effects of the pandemic and a host of other events have led to an awakening – highlighting the responsibility of companies to understand and respond to social shifts. Behind this responsibility lies the capability within organizations to govern the business with methods that create agility to alter course due to market shifts but also discipline to address environmental and social concerns. These concepts are also not new. Corporate social responsibility (CSR) has been on the table for several years. But this new world has shined a spotlight on the ESG issues. The issues hit the bottom line with customers, investors and even an organization's own employees watching closely. Being investor and climate driven, ESG is fundamentally a forward-looking Integrated Risk Management (IRM) approach to discern which companies are likely to thrive and which are likely to decline in a world growing in environmental and social uncertainty. ESG is more concentrated on the ability to sense and anticipate what is needed to prosper without doing harm to people and planet than it is about backwards looking control frameworks.

Reducing Carbon footprint in Steel Industry

Plant	CO2 emission Intensity (Apr'25-Jan'26) tCo2/TCS	CO2 emission Intensity FY25 tCo2/TCS
SAIL	**2.54**	**2.54**
BSP	2.55	2.62
DSP	2.50	2.48
RSP	2.54	2.53
BSL	2.61	2.62
ISP	2.41	2.36

Steel is one of the core pillars of today's society and, as one of the most important engineering and construction materials, it is present in many aspects of our lives. However, the industry now needs to cope with pressure to reduce its carbon footprint from both environmental and economic perspectives. As per Ministry of Steel, GOI, the iron and steel industry globally accounts for around 8 per cent of total carbon dioxide (CO_2) emissions on an annual basis, whereas in India, it contributes 12 per cent to the total CO_2 emissions. Thus, the Indian steel industry needs to reduce its emissions substantially in view of the commitments made at the COP26 climate change conference.

India’s steel sector accounts for about 12% of India’s carbon dioxide (CO2) emissions, with an emission intensity of 2.55 tonne of CO2/tonne of crude steel (tCO2/tcs) compared with the global average emission intensity of 1.85 tCO2/tcs. The steel industry is responsible for around 240 million tonnes of CO2 emissions annually and we expect this to double at an exponential rate by 2030, considering the Indian government’s infrastructure development targets.

There are multiple technology pathways that could help in the transition from traditional methods to low emission intensity technology like green hydrogen, renewable energy, carbon capture, usage and storage technology with Blast Furnace (BF)/Basic Oxygen Furnace (BOF) or Direct Reduced Iron (DRI)-Electric Arc Furnace (EAF), scrap-based Electric Arc Furnace (EAF) etc. Of these technologies, the green hydrogen-based route is the cleanest method of producing steel. However, green hydrogen is expensive and investing in the technology could render steelmakers uncompetitive as they sell a highly commoditised product.

The Ministry of Steel is committed to Net-Zero target by 2070. Towards this, in short term (FY 2030), reduction of carbon emissions in steel industry through promotion of energy and resource efficiency as well as renewable energy is being focused. For the medium term (2030-2047), utilisation of Green Hydrogen and Carbon Capture, Utilisation and Storage are the focus areas. For long term (2047-2070), disruptive alternative technological innovations can help achieve the transition to net-zero. For this purpose, Ministry of Steel is continuously engaging with various stakeholders.

Steps taken by MoS, GOI for promoting decarbonization in steel industry include:-

(i) Taxonomy for Green Steel (under development) – Working toward defining and certifying “Green Steel” to enable premium markets and green procurement.
(ii) Steel Scrap Recycling Policy, 2019 enhances the availability of domestically generated scrap to reduce the consumption of coal in steel making.
(iii) Ministry of New and Renewable Energy (MNRE) has announced National Green Hydrogen Mission for green hydrogen production and usage. The steel sector has also been made a stakeholder in the Mission.
(iv) Carbon Capture, Utilisation and Storage (CCUS) push – Supporting pilot projects and R&D for CO_2 capture in integrated steel plants.
(v) Motor Vehicles (Registration and Functions of Vehicles Scrapping Facility) Rules September 2021, shall increase availability of scrap in the steel sector.
(vi) National Solar Mission launched by MNRE in January 2010 promotes the use of solar energy and also helps reduce the emission of steel industry.
(vii) Perform, Achieve and Trade (PAT) scheme, under National Mission for Enhanced Energy Efficiency, incentivizes steel industry to reduce energy consumption.
(viii) The steel sector has adopted the Best Available Technologies (BAT) available globally, in the modernization & expansions projects.
(ix) Japan’s New Energy and Industrial Technology Development Organization (NEDO) Model Projects for Energy Efficiency Improvement have been implemented in steel plants.

Challenges in the Transition Journey of India

	World	India
CO2 Emission Contribution	8%	10–12%
Average Emissions Intensity	1.9 tCO_2/tCS	2.5 tCO_2/tCS
Use of Scrap	31%	21%
Natural gas Availability	High	Low
Quality of Ore	High	Low
Electricity Grid Carbon Intensity	Low	High

Green Electricity	**Availability of High Grade Ores & Scrap**	**Hydrogen Infrastructure**
Scale up of New Technologies	**CAPEX & OPEX**	**Green Steel Premium**
